

Unit 6, Lesson 7: Using Three-Column Tables of Equivalent Ratios

Benchmarks

MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.

Learning Targets

- ☐ I can use three-column tables to display equivalent ratios.
- ☐ I can solve problems using three-column tables of equivalent ratios.

6.7.1 Warm-Up: Brain Capacity

A human brain can store about 3,500 trillion bytes of data, while a cat's brain can store about 98 trillion bytes of data.

1. What is the brain storage capacity of human to cat?
2. About how many brains of cats would it take to store the same amount of data as the human brain?

6.7.2 Guided Instruction: Using Three-Column Ratio Tables

Ratios can be used to make part-to-part or part-to-whole comparisons. When comparing parts of a mixture to the whole, three-column ratio tables can be useful to represent the relationships between quantities.

1. Kiara is practicing her free-throw shot at basketball practice. She averages 4 made shots for every 2 missed shots. The table shown can be used to represent this situation.

Number of Shots Made	Number of Shots Missed	Total Shots Taken
4	2	

- a. When determining this rate during practice, Kiara missed 8 free-throw shots. Complete the second row of the table to show how many shots she made and how many total shots she took.
- b. How was each value in the second row determined?
2. Kiara had the opportunity to take 3 free-throw shots in her most recent game, and her accuracy remained the same as in practice. Complete the third row of the table to show how many free-throw shots she made and missed in the game.
- a. How was each value in the third row determined?

b. Complete the statement.

For every 3 free-throw shots taken, Kiara makes ____ shot(s) and misses ____ shot(s) on average. The ratio of missed shots to made shots to total shots is ____ : ____ : ____.

3. Three-column tables can also be useful when comparing mixtures that contain more than two parts.

Concrete can be created by mixing 1 part cement with 2 parts sand and 3 parts gravel. These dry ingredients are first combined before being mixed with water.



a. Complete the table to determine equivalent mixtures of concrete. Show how the values in each row are determined using arrows and by indicating the factor used.

Cement	Sand	Gravel
1	2	3
		1
6.5		
	10	

b. Marc uses one-gallon buckets to measure out the quantities needed in his concrete mixture. How many gallons of dry mix will he have in one batch of concrete?

6.7.3 Guided Practice: Using Three-Column Ratio Tables

1. A particular shade of maroon paint calls for 5 cups of red paint to be mixed with 3 cups of blue paint. Complete the table to find the quantities needed for different batches of paint.

Red Paint (cups)	Blue Paint (cups)	Maroon Paint (cups)
5	3	
	12	
		2
2.5		
		80

- a. If the mixture in the first row represents one batch of maroon paint, how many batches are in 80 cups of maroon paint?
- b. Explain how this relates to the factor used to complete the last row in the table.
2. There are 16 cups in one gallon. Use this information and the table below to determine how many gallons of paint the 80-cup mixture of maroon paint makes.

Cups	Gallons

3. In April, Kyla went for a walk on 4 days for every 1 day that she did not walk. There are 30 days in April. Use this information to complete the ratio table.

Walking Days	4	
Non-Walking Days	1	
Total Days		30

Complete the statement.

Throughout the year, Kyla continued walking 4 days for every 1 day that she did not walk. Based on this, in November, Kyla walked_____ of the 30 days.

6.7.4 Your Turn!

1. Bronze is a type of metal used to make many products such as ship propellers and parts of musical instruments. Bronze is made by mixing copper and tin in a ratio of 8.8 to 1.2. Complete the table where each quantity is measured in kilograms (kg). Then, answer the questions that follow.

Copper (kg)	8.8		
Tin (kg)	1.2		1
Bronze (kg)		25	

- a. How many kilograms of tin are needed to make 25 kilograms of bronze?
- b. To make bronze, how many kilograms of copper need to be melted per melted kilogram of tin?
- c. Draw an arrow to connect each question above to where the necessary information to answer can be found in the ratio table.

6.7.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed class that summarizes today's lesson.
